



THE PHILOSOPHY OF ADINKRA SYMBOLS IN ASANTE TEXTILES, JEWELLERY AND OTHER ART FORMS



Samuel Baah Kissi¹⁺
Peggy Ama Fening²
Eric Appau Asante³

^{1,2}Lecturer, Department of Industrial Art, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana.

¹Email: kissisamuelbaah@yahoo.com Tel: (+233)0244989869

²Email: amafening@yahoo.com Tel: (+233)0242606370

³Lecturer, Department of General Art Studies, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana.

¹Tel: (+233)0208163156



(+ Corresponding author)

ABSTRACT

Article History

Received: 14 January 2019

Revised: 22 February 2019

Accepted: 26 April 2019

Published: 1 July 2019

Keywords

Adinkra
Symbols
Philosophy
Stamping
Asante.

Adinkra, an iconographic writing which is a representation of the philosophy of the Asante people which is also understood by the people need to be preserved for generations to come. This research addresses the symbolism and philosophical presentations in the Adinkra symbols to the history of the Asante people. Qualitative research design (descriptive) was used to gather data. Various traditional printing studios and metal working studios were visited as well as review of some relevant literatures to the study. In building capacity for national development in relation to the philosophy of a people (Asante), various traditional working methods for production were evaluated, the research outline the systematic procedure both used in the traditional metal working industry for the production of jewellery with emphasis on traditional lost wax casting and that used in the traditional textiles industries in extracting dyes for use in fabric stamping. The paper also suggest possible improvements that can be either in the area of production process or in finishing the product and other alternative methods of using the Adinkra symbols in other art forms to preserve the symbols for posterity and for sustainable national development. Interpretation and concept development of some of the symbols with respect to the philosophy and history of the Asante people were discussed.

Contribution/ Originality: This study contributes in the existing literature on Adinkra symbols. It discusses the concept behind the symbols and tries to interpret the meaning of the symbols in relation to the philosophy of the Asante people. This paper further tries to bring a conclusion to the debate as to the origin of the symbols by analyzing various literature contributions which mostly turn to represent all Akans with Asante. The origin of the symbol is found with the Akans of which the people of Asante are a subset. Although the symbols have been with the Akans for long, the study suggest a way forward on how these symbols can be preserved for posterity without losing their philosophical trend with the Akan communities.

1. INTRODUCTION

Symbols and signs have been with man since the cave man's era as far back when Africa was referred to as the "Dark Continent" by the West. Symbols can be considered as ideographical representation of man's character and ideas which makes understanding much easier. Many cultures all over the world have different writing systems or symbols that convey messages and can be identified with the people. Numerals and Chinese characters are all forms

of globally accepted ideograms. Unfortunately many of these writing systems are unrecognized because they do not measure up to what some renowned Western linguists refer to as 'real' or 'true' scripts. Ong [1] and DeFrancis [2] define real and true scripts as writing systems that are based on phonetic systems. Therefore suggesting that writing systems that are not phonetic are inferior or in a developmental stage. Arthur [3] opines that, "pictographic, ideographic, and phonographic systems of writing do not represent inevitable stages in the development of writing as no direct evolutionary line can be drawn from pictographic to the phonographic system. However, critically analyzing the Asante's Adinkra writing system, it can be said that phonetic writing systems should be the basis of what is considered as "real writing", and that writing systems should be defined on how they are used to communicate within their respective cultures. This paper seeks to deliberate on Adinkra symbols and how they represent the history and philosophy of the Asante people of Ghana as well as suggest possible ways by which the symbols can be preserved. Development in all aspect of human life becomes relative and insignificant when it is not sustainable. In building capacity for national development in relation to the philosophy of a people (Asante), various traditional working methods for production were evaluated and possible improvements made either in the area of production process or in finishing the product to maximize marketability. Also channels of acquisition of knowledge should not only be restricted through a formal means of learning, but rather be a play mat displaying formal and informal levels with more emphasis on practiced based learner centered form of teaching.

1.1. Brief history of Adinkra

Adinkra which in Akan language means "farewell" or "goodbye" is an ideographic writing which represent ideas and teach valuable lessons. Moreover Adinkra symbols are visual metaphors because of their ability to enhance understanding by providing knowledge. Arthur [3] describes Adinkra as a coded system that induces meanings which preserve and present aspects of the belief, philosophy and history of the Asante people of Ghana. Adinkra symbolism has a name and meaning derived either from proverb, a historical event, human attitude, animal behavior, plant life, forms and shapes of inanimate and man-made objects [4]. The origin of the Adinkra symbols as stated by various authors and anthropologists sounds a bit controversial. The Asante's and Adinkra symbols are inseparable just as Bonwire and Kente are synonymous. Autry [5] stated that, Adinkra is believed to have been originated from Gyaman around the 19th century, a former kingdom in Côte D'Ivoire. Adinkra was the name of the King of Gyaman, who was captured in battle by the Asantes for making a replica of their "Golden Stool" which represents their power and cohesion.

History has it that, Nana Kofi Adinkra, the King of Gyaman wore a patterned cloth which was interpreted as a way expressing his sorrow. Some royals were captured by the Asante's in battle and it's believed they thought the Asantes how to print the symbols in clothes. This is a theoretical truth which a lot of authors believe but how far this is true has yet not received a concrete basis since the philosophy and history of the individual symbols can mostly be traced to the Akans better than any other culture. There are other several conflicting ideas to the origin of the symbols. According to Ratray [6] before the Asante-Gyaman war in 1818 these symbols were already in existence as introduced by the Mohammedans through trade. These traders had already introduced amulets that had designs similar to the Adinkra symbols. This implies that perhaps the symbols were copied from the Mohammedans through the Northern trade. Thomas Edward Bowdich has also given different perspective of the date in which Asantes first came into contact with Adinkra symbols in relation to the Gyaman war. Bowdich has speculated that he had collected a piece of Adinkra cloth in 1817 [7].

According to Danzy [8] Adinkra cloth came with the Asantehene's golden stool. In the history of the Asantes, there is a belief that Okomfo Anokye, the greatest chief priest ever lived in the history of the Asante called upon the heavens to bring down the golden stool, an artifact that came to symbolize the power of the first King of the Asante nation, Osei Tutu as well as the power of every succeeding Asantehene.

It is believed that Adinkra cloth was folded on top of the golden stool, brought down from the heavens; which means that the symbols were divinely given and not copied from any other culture. There are no existing facts to support these theories but authors cling to the fact that these symbols were obtained from the King of Gyaman simply because his name was Adinkra. Upon analyzing the various accounts as to the origin of these symbols, it can be deduced that the Adinkra symbols originated from the Akans and not the Gyamans as many authors speculated.

History has it that series of war forced the Akan people to split into clans and all settled in different corners of Ghana and some settled in Côte D'Ivoire whose descendant is Nana Kofi Adinkra. More over the name of the then King of Gyaman and the names given to the symbols are all in "Twi" which is the official dialect of the Akans, indicating that the symbols have Akan origin. Also considering the account of Bowdich and Danzy, it can be deduced that the Gyaman war which dates 1818 [9] came about years after the golden stool was chanted down by Komfo Anokye. This indicates that if the account of Danzy is to be taken as true then Adinkra was already with the Asantes before the War as Bowdich's encounter with an Adinkra artefact in 1817 confirms the fact that Adinkra art was with the Asante's even a year before the war. Also straining the yarns in Rattray's account with the philosophy of the symbols, the Adinkra symbols are pregnant with Akan history and proverbs and has nothing to do with history in Northern Ghana. There are still not any tangible artefacts or historical items which can be found in the Northern part of Ghana to substantiate the fact that Adinkra begun in the North. Adinkra cannot be said to belong to or founded by the Asantes alone rather all the Akan people in Ghana since their philosophy and history can be traced to almost all the Adinkra symbols.

2. THE ADINKRA IN CLOTHES

African art are not done for just art sake but for a purpose and they can be identified with the people. Kente, Adanudo and Adinkra clothes are some of the very few traditional cloths associated with the people of Asante which convey the thought and feelings of the wearer. Adinkra cloths were traditionally worn by royals and spiritual leader for special ceremonies, but in recent times, it is worn by non-royals and even tribes afar for any befitting occasion [10] and African History [11].

Until the introduction of the symbols in cloth printing, Adinkra symbols were used in architecture such as in the shrine of Yaa Asantewaa at Ejisu in the Asante Kingdom as well as on ceremonial article such as swords and consecrated items such as the stools used at the shrine and at Asantehene palace at Manhyia. History has it that a hunter named Kojo from Nana Boakye family discovered the printing dye locally called "Adinkra Aduro" when on one expedition had his cloth stained with fluid from "Badie" tree in his quest to cut down the tree. It's said he made several attempts to wash off the tree stain but could not and it kept darkening by the day. After sometime, it is recorded he later went for the bark of the tree and deliberately stained all of the dress and this act triggered a huge family venture that is practiced by the people of Ntonsu till date [12].

Series of production activity takes place before one can obtain a complete Adinkra cloth, such steps include the carving of the design unto a calabash Figure 1 and then erecting the carved design unto a palm front pedestal. The bark of "Badie" tree is collected from the farms and soaked for in water a week as can be seen in Figure 2, and then pounded, extraction of the dye from the tree back is done by boiling. The stock collected is again boiled until a thick paste residue is achieved Figure 2. Stamping can be done on either strips of plain woven Kente Figure 5 which are latter joint together into full cloth or machine woven fabric which is died and then stamped Figure 6. Stamping is done by dipping the carved design into the dye paste Figure 3 and then pressing the design unto the fabric Figure 4.



Figure-1. Carved Adinkra Symbols.

Source: Field Research 2016.



Figure-2. Soaked "Badie" Tree Bark.

Source: Field Research 2016.



Figure-3. Printing paste "Adinkra Aduro".

Source: Field Research 2016.



Figure-4. Adinkra Stamping.

Source: Field Research 2016.



Figure-5. Stamped woven Kente.

Source: Field Research 2016.



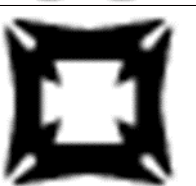
Figure-6. Stamped Machine made fabric.

Source: Field Research 2016.

2.1. Some Sample Adinkra Symbols and their Meaning

Adinkra symbols appear on some traditional Akan gold weights, cloth and linguist staff. Some are carved on stools for domestic and ritual use. There are several categorizes of Adinkra symbols but this research shall be limited to five categories, namely;

- Educational symbols
- Spiritual / Religious symbols
- Political symbols
- Funeral symbols
- Social symbols

Category	Symbol	Name in Akan/English	Symbolic Meaning
Educational Symbols		<i>Ntesie</i> I have heard and kept it	A symbol signifying the need for imbibing knowledge, information, wisdom and prudence.
		<i>Sankofa</i> Going back to my beginnings	Going back to my roots, to undo or correct mistakes, to build and plan a better future.
Political symbols		<i>Adinkrahene</i> King of the adinkra symbols	Symbolizes greatness, charisma and leadership
		<i>Ohene Anirwa</i> The King's eyes	Symbol of Vigilance and wariness. The king has eyes every where
		<i>Akofena</i> Sword	Symbol of authority and gallantry. A great retiring warrior deserves his honour
Spiritual / Religious symbols		<i>Gye Nyame</i> Except God	Symbolizes supremacy, Omnipotence of God the Divine Creator
		<i>Nyame Biribi Wo Soro</i> God there is something in the heavens for me	Symbol of hope
Funeral symbols		<i>Owuo atwedie</i> Ladder of death	Death the ultimate fate of man. Mortal nature of man. Fate
Social symbols		<i>Nkonsonkonson</i> Chain links	Human relations, Unity
		<i>Fihankra</i> House	Safety /security in a home

2.2. Aesthetic Elements and Values

Till date, Africans have developed and are using both oral and visual images to convey, and store their emotions and attitude. These images are used symbolically to communicate knowledge and teach values and morals.

A symbol help in giving clear information about the environment, feelings of people, and hide and protect some important secrets. Ghanaian symbols known as Adinkra are visual symbols that represent concepts and are used in fabrics, on walls, logos and can be found everywhere.

There are several bright colour cloths which are used for Adinkra projects. There are also dark and dull colours of dark-brown, black and brick red made up of “kuntunkuni”, “birisi” and “kobene” cloths. The Asante people’s aesthetic response to black is sadness and hopelessness, and red is blood or death. That is why “kuntunkuni” and “kobene” are worn for funerals and to show sympathy.

Aesthetically the symbols are printed in cloths, carved in stools and on building for beautification purpose side the message they carry, a typical example is the architectural designs at Yaa Asantewaa shrine in Ejisu [Figure 7](#).

The Adinkra symbols are all formed from lines, dots and shapes. Majority of the symbols are rounded in form comprising of curves, horizontal and vertical lines which are inter twine and intersect at an angle giving them the unique features that cannot be seen in any writing style. It cannot be said to have followed or copy Chinese writing style.

Some of the symbols such as Akofena (sword) and the symbol of gun and tortoise were all developed and design from the immediate environment to which man owe a great sense of deal as humans and the design of the sword is not foreign but a traditional one which is easily identified with and by the people.



Figure-7. Adinkra symbols in Architecture.
Source: Field Research 2016.

3. ADINKRA IN ASANTE JEWELLERY

Adinkra symbols have been with Asante for centuries primarily in the area of fabric printing and a minimal number in their wood carvings. However, the origin of the use of adinkra symbols in metal art such as jewellery cannot clearly be identified but its believed to have begun around the 1700`s years before the demonetization of currency from gold dust to paper notes and coins where lost wax casting technique was the primary metal working technique used in the production of jewellery for the “Asantehene” around Ensuasi a suburb of the Asante Kingdom. This technique of metal working kept the artisans in serious business not because for the sole purpose of jewellery making but the production of gold weights used on balance scale for trade around the 1700`s. The artisans developed ideas or took inspiration from myths, proverbs and already existing adinkra symbols which were primarily used in clothes. There were no preliminary sketches made. Whatever idea the artisans had was first tried out in wax as sample work. The artisans use manual-traditional technology for the production of the artefacts. They work in open spaces, court yards and in front of their houses. An artisan workshop comprises a simple wooden shed, under which the artisan works, a working table and chair, a modified crucible and a container for keeping tools. The

furnace is sighted a few meters from the shed. The artisans' tools included the following; weighing scale, mortar and pestle, sieve, plastic containers and a heat source. They use local materials such as bee wax, clay, charcoal, palm fibers and cow dung. The initial funds required for the establishment of a workshop is therefore not very high [13, 14].

The artisans basically use the lost wax casting method for the production of their artefacts. The designs are beautifully sculpted out of soft beeswax Figure 8. Then they are dipped into a mix of very fine charcoal and water to give a coating which holds all the fine details Figure 9. This is then covered in a mould of coarse palm fibers and clay Figure 10. The mould is heated and the wax is poured out. Scrap brass is then melted in a crucible and poured into the mould bearing cavity of the intended pattern Figure 11. This allows the metal to melt and drain down into the spaces left by the wax. The mould is taken out of the kiln and after cooling broke open to obtain metal mould. Charcoal is removed from inside hollow mould and metal is filed down to get smooth finish which is then polished using lemon juice, an iron brush and water. The artefacts produced are visually aesthetic, unique, and simple [15]. They carry messages which represent the content of their culture that is knowledge, views, values of life orientation, and the belief system. Each art form symbolizes the culture of the people as represented in Figure 12 and Figure 13.



Figure-8. Wax forming.

Source: Google.



Figure-9. Coating in charcoal clay mixture.

Source: Google.



Figure-10. Coating with clay and palmfiber.

Source: Google.



Figure-11. Casting process.

Source: Field Research 2016.



Figure-12. Cast Adinkra Pendant.

Source: Field Research 2016.



Figure-13. Cast Adinkra Signet Rings.

Source: Field Research 2016.

4. METHODOLOGY

According to Key [16] qualitative research is a generic term for investigative methodologies which can be broken down further into the following sub-headings: naturalistic, field or observer research, ethnographic and anthropological research with emphasis on interaction between variables and detailed data gathering through open ended questions that provide direct answers.

This research is qualitative and used descriptive research methods for data collection

According to Babbie [17] descriptive research is the accurate and systematic description of data and characteristics about the population or phenomenon being studied. Semi-structured interviews were employed as a means of gathering data for this research. Face to face interaction was held with craftsmen and resource persons so that the needed facts could be obtained and documented. Various related literatures were reviewed and results analyzed and synthesized.

The nature of the research topic demanded that several traditional Adinkra Printing centers be visited at Ejisu, Manhyia, Bonwire, Ntonsu and Kumsi Culture Center to allow the researcher examine, analyze and describe various production processes and products that make use of Adinkra symbols, and document all.

5. DISCUSSION

It is an undisputable fact that Adinkra symbols are visual representation of not only Asante's but Akan history of which Asante form part of and their philosophy. Several examples have been stipulated below to substantiate this claim.

5.1. AKUFENA

Which represents authority and power when carefully analyzed give the history and philosophy of the Asante people. The design chosen for that particular symbol is the traditional sword. The sword in its self is not ordinary and cannot be identified with the Northerners in Ghana nor other cultures other than the Akan tribes of Ghana. One may ask why the sword to represent authority and not a gun or currency (money)? More over even if the designers wanted a weapon other than gun but within cutting tools perhaps they could have gone for pen knife or even machete and not a traditional sword. Again if the choice was for a sword may be they could have chosen the sword design used by the Queen of England since they are our colonial masters? With all this questions unanswered, it brings us to the history and philosophy of the symbols. The Akufena was used in war, hence the name Aku-Fena which literally means War-Sword. In the very old days before the introduction of guns by the West into Africa, swords, bow and arrows and spear were the weapons for battle. It was used to wage war and also during battle when a king is captured and he is beheaded, his head was tied to the sword and kept by the victors, the captives then pay allegiance to the king who now has the sword and the head of the past king and that shows he has

authority over their lives. It was used for swearing in because it represent the freedom of his subjects and he has to protect it with his life and if by chance he is defeated in a battle and beheaded all his subjects becomes slave to whosoever has his head.

Hence the sword stands for the freedom of the people and whosoever has power over their freedom has authority and as a symbol of permission to lead. This is the reason why even till date it is used for swearing Presidents into office, chiefs and leader of various organizations.

5.2. GYE NYAME

Which literally means Except God and represents supremacy, Omnipotence of God the Divine Creator. Looking at the symbols it can be seen that a lot of deep thinking went into design and it brings out how the Asante think of God and who they believe him to be. For instance, two main shapes can be seen in the design being: round balls and two curved poles like a horn. The balls seen in the *gye nyame* design is similar to that in the *Ohene Anirwa* design which means that God is also perceived as a king who also sees everything. The arrangement of the balls too when one draws line to follow the pattern, the outcome was similar to the *Nkyemkyem* symbol which indicate how deep God is that no one can fully map out Gods ways and strategies he deals with man. Also the curved poles on each side like a horn is also similar to the *Akobon* design which also implies that God is known to them as a strong force that none can recon with. The horn which also stands for judgment and strength with each on both sides pointing up and down shows that non can escape God`s judgment. Lastly, drawing straight line through the Gye Nyame design from the top to bottom, it gives a mirror reflection which has been turned upside down representing the unchanging nature of God as believed by the Asante`s.

5.3. NTISIE

Meaning, I have heard and kept, which signifies the need for imbibing knowledge, information, wisdom and prudence. The design is made of a distorted circle thus elliptical shape and a dot. The position of the dot in the circle seem similar to an open mouth with the tong. Then it is not by mistake that the dot is found within the circle and not outside. There is also the proverb in Akan which says “one head cannot take counsel” and also “two heads are better than one”. May be this could have informed the designers of the symbol to attach four circles as seen in the symbol. Also another loud thinking can be that the circles represents the ears and the dot as the hearing drum, and each individual has two ears and four circles represent two people reasoning together. It also shows that the Asantes mostly regard quiet people as wise because it is said that silence is gold. It also confirms the Akan saying that, great tasks are resolved in silent and refrain from rebutter keeps the stick to sleep. The Asante`s also believe that a king does not dwell on hearsay.

There are also other Adinkra symbols aside the ones discussed in this report that represent the philosophy and history of the Asante. One common example is the symbol of snail, tortoise and gun, (*EKA AKTEKYEDI NI NNWA NKO NNKA ETUO NTO KWAYEMU*) meaning, if left for the tortoise and snail there will be no gun shot in the forest, which signifies peace. The symbol tell the history of the Asante in terms of occupation as they were one hunters and also it relate to the numerous battles they fought. But the designers chose tortoise and snail around a gun because the use of a gun is not needed in hunting for this animals and also teaches the lesson of patience and being hopeful. The animals are slow but with patience and hopefulness can achieve everything. As there is an adage in Akan which goes by “patience is a sign of victory”.

6. RECOMMENDATION

Adinkra is a strong component of Akan culture and Ghana as a whole. It represents a traditional culture that is gradually diminishing because of the growing global capitalist economy. Although the meaning of almost all the symbols are not understood by all Ghanaians, yet these symbols are worn by many and used as institutional logos

and on institutional and public buildings an example is Golden Tulip Hotel both in Kumasi and Accra, University of Ghana buildings [18]. To mention but a few are: the president and vice-president's ceremonial state seat, speaker of parliament's robe and the state sword which are all clouded with Adinkra symbols.

According to Frimpong, et al. [19] and Akakpo [20] Adinkra symbols are not really recognized for the messages that they communicate but rather their aesthetic features that appears on a variety of art mediums.

The Adinkra symbols can be manipulated and modified without losing its basic features and used in other disciplines than just in Architecture and traditional funeral clothes printing and be used in T-shirt printing, product branding, sculpture, leather, ceramic, jewellery and metalsmithing industry. The designer can stop using the symbols in isolation, and only representing the symbols and their meanings more in only books rather can be coined with its meaning to form one icon which will showcase the meaning and symbol as one design. Jewellery that have been pierced in Adinkra symbols can be exported for foreign earnings, and also help other cultures develop the taste for symbols and not necessarily the meanings they convey.

In as much as the symbols have become a major identity with the Ghanaian people as a whole and not only to the Asantes from Akan, the niche market created through the application of these symbols in different art forms has not been harnessed fully. Finishing of the works of art must be improved appreciably. In this era of globalization of trade and services where complex machinery have been developed and imported artefacts are competing aggressively with the locally produced contents, packaging as well as surface finishing techniques must be developed through the use of available local mechanism so as to bring up uniqueness in the products produced locally. It is an undisputable fact that there are limited machines available yet much can be done with the simple tools available than the blunt finishing and poor packaging executed by local artisan.

Again, jewellers should not restrict the use of the symbols to only rings, necklaces, chains, earrings, bracelets but should also be producing belt-buckles, wrist watched, sun glasses, hair pins and the likes. This will help open up their customer size and in effect influence their income.

Funding: This study received no specific financial support.

Competing Interests: The authors declare that they have no competing interests.

Contributors/Acknowledgement: All authors contributed equally to the conception and design of the study.

REFERENCES

- [1] W. J. Ong, *Orality and literacy*. New York: Methuen: The Technologizing of the Word, 1982.
- [2] J. DeFrancis, *The Chinese language: Fact and fantasy*. Honolulu: University of Hawaii Press, Designs and Publications, 1984.
- [3] G. F. K. Arthur, *Cloth as a metaphor: Reading the adinkra cloth symbols of the Akan of Ghana*. Legon: Centre for Indigenous Knowledge Systems, 2001.
- [4] K. Richardson, "Adinkra history." Available: <http://www.probusportmelbourne.org.au/>. [Accessed 7 December 2015], 2012.
- [5] A. Autry, *Farewell medicine. African Diaspora ISPs. pp: 81*, 1999.
- [6] R. S. Rattray, *Religion and art in ashanti*. London: Oxford University Press, 1988.
- [7] Afroetic, "Adinkra symbols." Available: <http://afroetic.com/adinkra-symbols/> [Accessed 6th December, 2015], 2011.
- [8] J. Danzy, "Adinkra symbols: An ideographic writing system." The Graduate School at Stony Brook University. (Unpublishedtheses), 2009.
- [9] GMMB, *Asante traditional buildings*. Kumasi: Ghana Museums and Monuments, 1999.
- [10] J. O. Boadu, "Outburst: Adinkra as a good communication tool in Ghana." Available: <http://juliusboadu.blogspot.com/2011/11/normal-0-false-false-false-en-gb-x-none.hhtml>. [Accessed 8 December 2015], 2011.

- [11] African History, "Adinkra symbology and its origins." Available: <http://africanhistory.about.com/library/weekly/aaAdinkra.htm>. [Accessed 6 December 2015], 2015.
- [12] K. Boateng, *History and origin of adinkra symbols*. Ghanaacul: Ghana Cultural Politics, 2015.
- [13] R. S. Rattray, *Religion and art in ashanti*. London: OVP, 1923.
- [14] P. A. Fening, S. B. Kissi, and E. A. Asante, "Preservation and enculturation of the metal smithing industry in Ghana: The case of traditional brass casting in 'Krofrom' village," *The International Journal of Humanities & Social Studies*, vol. 4, pp. 150-155, 2016.
- [15] S. B. Kissi, P. A. Fening, and D. Adom, "Beeswax in asante castings: The tthen, now and the way forward," *Journal of Arts and Humanities*, vol. 5, pp. 07-22, 2016. Available at: <https://doi.org/10.18533/journal.v5i10.984>.
- [16] Key, "Qualitative research design." Available: <https://www.google.com/search?q=google%20search>. [Accessed 4th September, 2009], 1997.
- [17] L. B. Babbie, *Qualitative research methods for the social science* Boston: Pearson Education, 1989.
- [18] A. K. Quarcoo, *The language of adinkra patterns*. Legon: University of Ghana, 1994.
- [19] C. Frimpong, B. Asinyo, and A. Amankwah, "Contemporary trends in Adinkra cloth production: Design, printing technique, base fabric and printing paste (Dye)," *International Journal of Fiber and Textile Research*, vol. 3, pp. 43-48, 2013.
- [20] E. Akakpo, "Adinkra symbols." Available: <http://www.akakpo.com/adinkra-symbols/>. [Accessed 8 December 2015], 2015.

Views and opinions expressed in this article are the views and opinions of the author(s), Journal of Asian Scientific Research shall not be responsible or answerable for any loss, damage or liability etc. caused in relation to/arising out of the use of the content.

© 2019. Notwithstanding the ProQuest Terms and Conditions, you may use this content in accordance with the associated terms available at <http://www.aessweb.com/journals/5003>